Grade 10 – Term 1
Interactive Decision Webpage: C10 Practice

Evaluated Criterion – C10

Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, objectives, and constraints.

1 Your Assignment

Create and deploy an **individual interactive webpage using HTML, CSS, and JavaScript** that analyzes a realistic economic or financial decision under uncertainty. Use the C10 reference dashboard as your functional model: decision context and editable payoffs, automatic summary, criterion decisions, and staged filtering with a flow diagram. Add your own interpretation, recommendation, references, and AI development record directly to the webpage.

This project synthesizes C1–C9: identifying alternatives, states, and consequences (C1); organizing payoff tables (C2); calculating profit as revenue minus costs (C3); applying Maximax and Maximin (C4), Minimax Regret (C5), and EMV/Bayes (C6); examining subjective probabilities and sensitivity (C7); comparing criteria (C8); and constructing and interpreting staged filtering trees (C9). **Calculating all four criteria and implementing filtering integrates earlier work. The central C10 task is to interpret the results and defend which criterion best fits your chosen situation.** Include this explanation on the webpage; no separate written report is required.

2 Prepare Your Project

1. **Choose a manageable decision.** For example, investigate a small business's stock purchase or an event's spending plan with uncertain attendance. Start with at least two feasible alternatives and two possible states of nature. Identify the decision-maker, objective, time period, risk attitude, and constraints, such as budget or acceptable loss. Explain exclusions of infeasible alternatives before comparing choices.
2. **Prepare comparable financial information.** Define mutually exclusive states covering the relevant possibilities. Investigate revenues and costs and explain how you derive net profit, net gain, or savings for each alternative–state combination. Use consistent currency, units, and time period; larger payoffs must mean better outcomes. Convert cost information into comparable net payoffs and explain the conversion. Negative payoffs represent losses.
3. **Separate evidence from assumptions.** Use verifiable information where possible, such as published prices, quotations, or historical records. Cite sourced facts where used; clearly label and justify assumptions and estimates. Future payoffs are estimates, not established facts. If probabilities are available, explain their source or estimation basis; do not invent probabilities or assume equal probabilities merely because they are unknown.
4. **Plan and build the webpage.** Review a previously worked C1–C9 example and the reference dashboard before implementing the sections below. Use HTML for content, CSS for readable layout, and JavaScript for controls and calculations. Give the project a descriptive title and include your full name.

3 Required Webpage Structure

3.1 Decision Context and Editable Payoff Table

Present the decision-maker, objective, alternatives, states of nature, constraints, currency, and time period. Allow users to edit alternative and state names, add or remove alternatives and states, and enter payoffs and state probabilities. Clearly label the table orientation: alternatives may be rows or columns, provided every calculation follows that orientation consistently. Label sourced financial information, assumptions, and estimates and explain how the starting payoffs were obtained.

3.2 Automatically Updated Summary

For every alternative, display EMV, maximum payoff, minimum payoff, and maximum regret. Include an inspectable regret table and brief explanations of the calculations. Let P_{ij} denote the net payoff for alternative A_i in state S_j, and let p_j be the probability of that state. Use

$$M_i = \max_j P_{ij}, \qquad m_i = \min_j P_{ij}, \qquad EMV_i = \sum_j p_j P_{ij}.$$

For each state, find its best payoff among *all original alternatives*, then calculate regret:

$$b_j = \max_i P_{ij}, \qquad r_{ij} = b_j - P_{ij}, \qquad R_i = \max_j r_{ij}.$$

Thus, R_i is maximum regret. Regret measures a missed opportunity relative to the best choice for that state; it is not necessarily a cash loss.

3.3 Criterion Decisions

Compare the complete original set of alternatives under each of the four criteria. Show **winners, rankings, deciding values, and ties**, and explain each criterion:

- **Maximax:** highest M_i; an optimistic approach pursuing the greatest possible upside. It ignores other outcomes and their likelihoods.
- **Maximin:** highest m_i; protects the worst-case payoff. It can overlook substantial upside.
- **Minimax Regret:** lowest R_i; limits the largest missed opportunity. It does not directly maximize profit or account for probabilities.
- **EMV/Bayes:** highest EMV_i; maximizes probability-weighted average money when justified probabilities and a risk-neutral monetary objective fit the context. Expected value is not a guaranteed outcome and does not protect against a large loss.

Explain why different priorities and information requirements can produce different recommendations. The first three criteria remain usable without probabilities; show EMV as unavailable when probabilities are missing or invalid.

3.4 Staged Filtering and Flow Diagram

Allow users to **create, reorder, and remove stages**. At each stage, let them choose any of the four criteria and either retain the best k alternatives or select the best. Higher values are better for Maximax, Maximin, and EMV; lower values are better for Minimax Regret.

- Start with the complete original set. Apply each stage only to the preceding survivors; eliminated alternatives must never reenter later in that sequence.
- Calculate all summary statistics and state regret benchmarks from the complete original payoff table. Keep them fixed throughout filtering; do not recalculate regret using only survivors. When inputs change, recalculate from the updated complete table and restart the sequence.

- Retain every tie at the cutoff, even if more than k alternatives survive. "Select the best" must retain all tied best alternatives. Require a positive whole number for k; if it exceeds the survivor count, retain all survivors.
- Show a step-by-step table with step number, criterion/action and relevant values, surviving alternatives, and alternatives eliminated at that stage. Step 0 must show the complete set. Identify the final survivors or tied final decision.
- Show a corresponding flow diagram connecting successive survivor groups and marking eliminations. The diagram and table must agree. If an EMV stage cannot run because probabilities are unavailable or invalid, explain why and do not show a misleading completed sequence; probability-free criteria must remain usable.

3.5 C10 Interpretation and Recommendation

Write your own connected explanation on the webpage, using complete paragraphs and the results for your starting scenario.

1. **Select and defend a criterion.** Explain which criterion best fits the available information, objectives, risk attitude, and constraints. Compare it conceptually with the other three. Choose based on the situation, rather than whichever rule gives your preferred alternative.
2. **Interpret the recommendation.** State the recommended alternative or tie and explain the deciding values in the original currency, units, and time period. Connect the outcome to the realistic economic or financial choice and the decision-maker's ability to bear losses.
3. **Distinguish the two types of output.** Individual criterion decisions compare all original alternatives. Staged filtering answers a different question determined by the order and rules of its stages. Explain your demonstrated sequence and its outcome separately; it does not by itself justify which criterion is most appropriate for C10.
4. **Discuss limitations and sensitivity.** Explain what your chosen criterion ignores and where evidence or estimates are uncertain. Test a plausible change in an important payoff assumption or, where justified, probability assumption. Describe whether and why the recommendation changes; keep any revised probabilities valid. This develops the sensitivity work from C7.

3.6 References and AI Development Record

Include a **References** section with the sources used for contextual claims and financial data. Cite them where used and provide enough detail to locate them: author or organization, title, date, and accessible link where available.

The deployed webpage must also have a visible section titled **AI development record** or **AI conversations used**, containing accessible hyperlinks to the AI conversations used in development. These links must open directly from the deployed webpage. Check their sharing/access settings; links available only to your own signed-in account are insufficient. You are responsible for understanding and verifying the calculations and explanations produced with AI assistance.

4 Mathematical Accuracy, Usability, and Testing

- Support negative payoffs and decimals. Validate missing or invalid inputs and show useful messages identifying what needs correction. A blank payoff is not zero; do not display misleading results from incomplete data.
- EMV requires justified state probabilities, each between 0 and 1, summing to 1. Display the total and explain any small numerical tolerance used for decimal arithmetic. **Do not invent probabilities or automatically normalize incorrect totals.** When probabilities are unavailable or invalid, label EMV unavailable while keeping Maximax, Maximin, and Minimax Regret usable with valid payoffs.
- Update summaries, rankings, ties, filtering steps, and the diagram when inputs or stage controls change. Apply a consistent tie rule, explain any numerical tolerance, and calculate with full precision before rounding displayed values.
- Provide readable tables, labeled controls, clear navigation, and a layout usable on desktop and mobile. Ensure wide tables and the flow diagram remain accessible on small screens.
- Verify calculations against a previously worked example: check payoffs, summary statistics, regret entries, criterion winners/rankings, and filtering survivors. Test edited names and values, added/removed alternatives and states, stage reordering/removal, negative values, decimals, ties (including a cutoff tie), blank or invalid inputs, and missing/out-of-range probabilities or totals different from 1. Confirm that invalid probabilities disable EMV without disabling

probability-free criteria. Include a brief verification note on the webpage naming the example and summarizing these checks.

5 Deploy and Submit

1. Prepare a **GitHub repository** containing the webpage's HTML, CSS, JavaScript, and any required assets. Ensure the teacher can access the repository and source files.
2. Publish the functional webpage using **GitHub Pages** or another accessible public/shared hosting link. Use these tutorials if needed:
   - Tutorial to download Git and push a project into GitHub.
   - Tutorial to host a website using GitHub Pages.
3. Open the **deployed URL** and check that controls, calculations, tables, diagrams, sources, and AI conversation links work. Check desktop and mobile usability and confirm shared links are accessible to the teacher.
4. Submit both the **deployed webpage URL** and the **GitHub repository URL**. Screenshots, source files alone, and PDF exports do not replace the deployed functional webpage.

6 Final Submission Checklist

- **Mathematics:** comparable net payoffs; correct summaries, regrets, rankings, ties, and probability validation; worked-example verification completed.
- **Functionality:** editable table; automatic updates; working stage controls; fixed original benchmarks; consistent survivor table and flow diagram; readable desktop/mobile layout.
- **C10 justification:** student-authored criterion selection, contextual recommendation, comparison, limitations, and plausible assumption-change analysis; individual decisions distinguished from filtering.
- **References:** sourced facts cited and distinguishable from justified assumptions and estimates.
- **Deployment and repository:** both URLs submitted; deployed functionality tested; repository and source files accessible to the teacher.
- **AI links:** visible AI development record with accessible conversation hyperlinks checked on the deployed webpage.